DATABASE MANAGEMENT SYSTEMS
[ENCS204]

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
[LAKSHITA KALRA](2401010117)

Course Teacher:
MISS MANSI KAJAL

School of Engineering and Technology

K.R.MANGALAM UNIVERSITY
Sohna,Haryana 122103,India
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LAB ASSIGNMENT-1

Aim:
To design ER diagrams for different database scenarios and convert them into relational schemas. To implement the schemas using SQL commands (DDL and DML) in a DBMS tool.

Description:
In this experiment, we studied how to design a database system.
First, we created ER diagrams to understand entities and relationships.
Then we converted those diagrams into relational schemas (tables).
After that, we created tables in MySQL and applied primary key and foreign key constraints. We also performed DDL operations like ALTER, DROP and DML operations like INSERT, SELECT, VIEW creation.

Procedure:

STEP 1: Open DBMS Tool

- Open MySQL Workbench

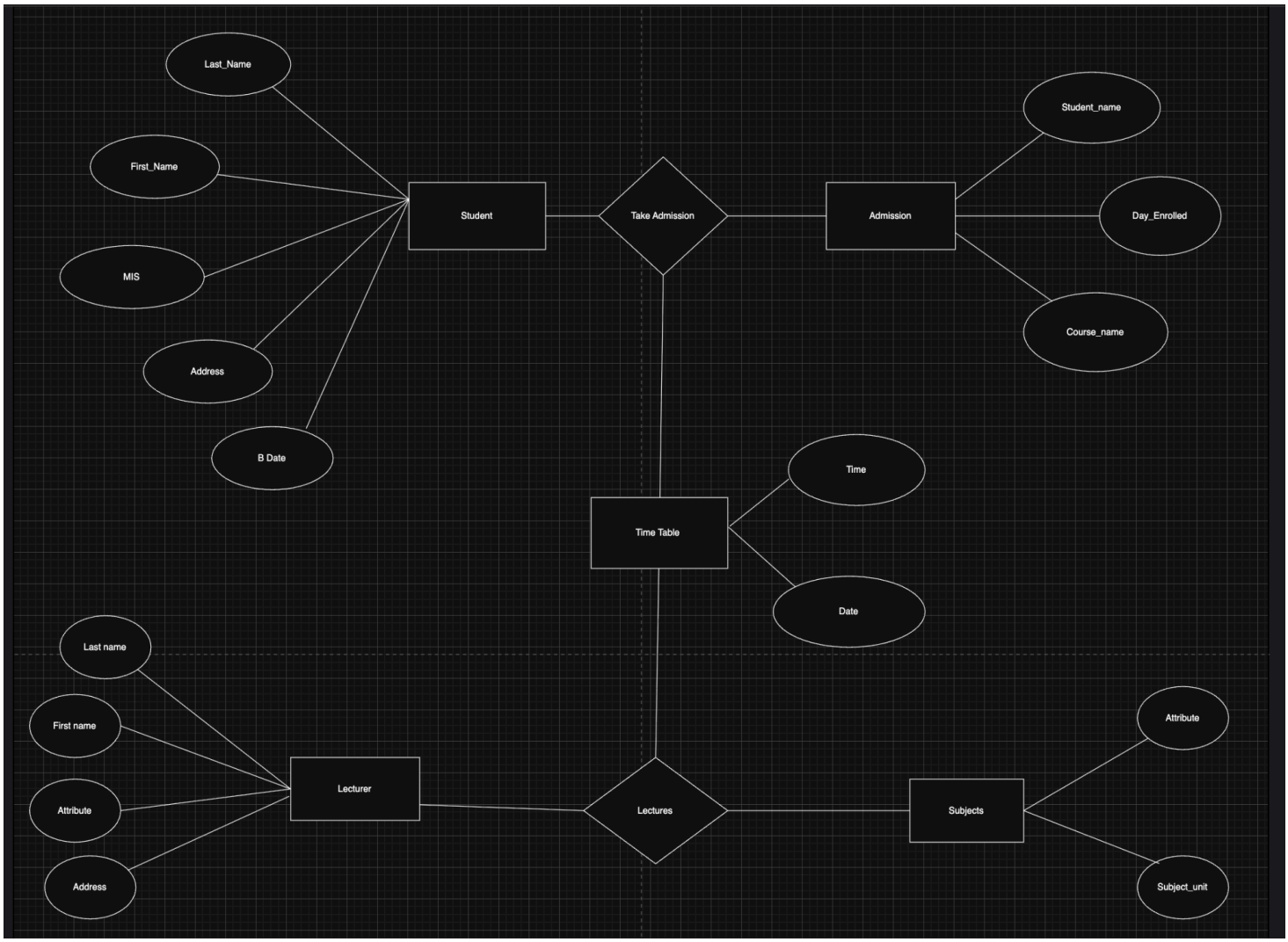
STEP 2: Create Database

```
CREATE DATABASE DBMS_Lakshita;
USE DBMS_Lakshita;
```

1	21:18:25	CREATE DATABASE DBMS_Lakshita	1 row(s) affected	0.0095 sec
2	21:18:25	USE DBMS_Lakshita	0 row(s) affected	0.00037 sec

PART A – COLLEGE DATABASE

STEP 3: Draw ER Diagram



STEP 4: Create Tables

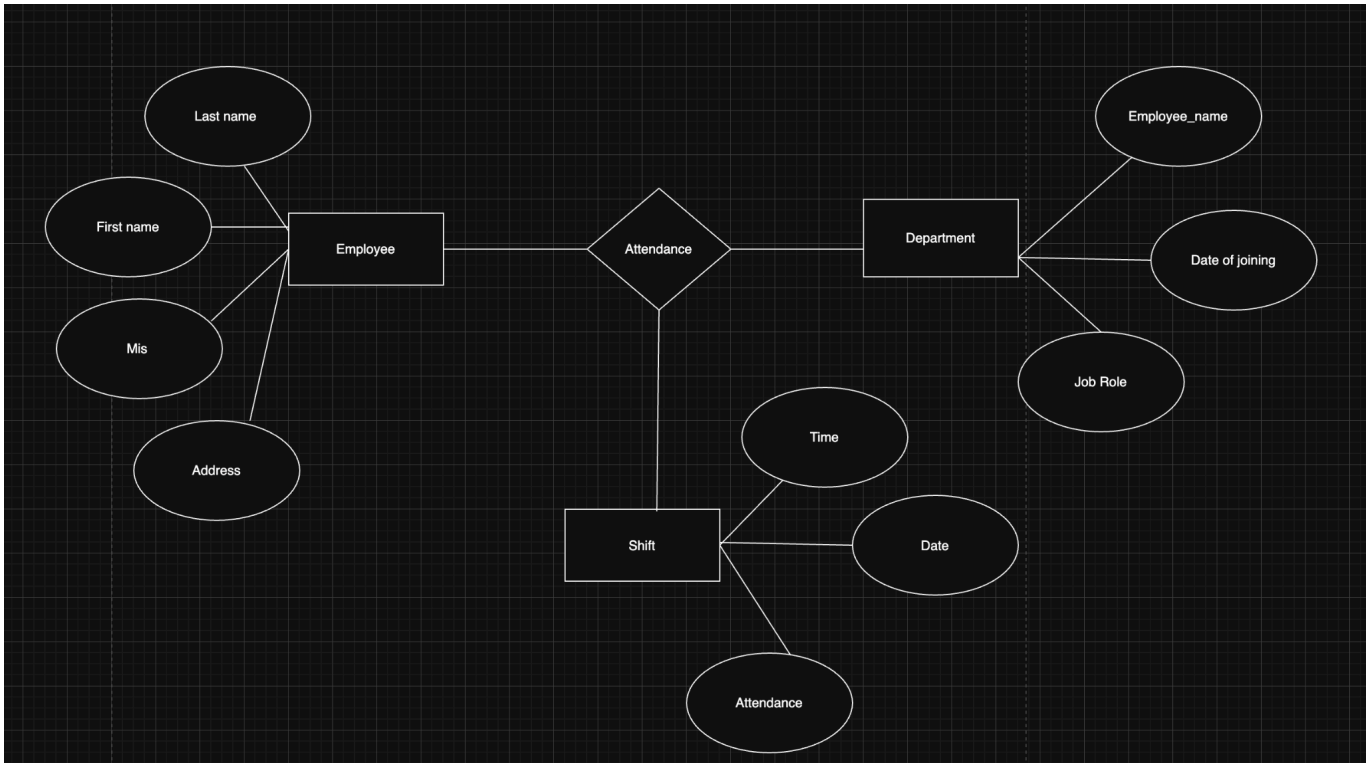
CREATE TABLE STUDENT(
USN VARCHAR(20) PRIMARY KEY,
SName VARCHAR(50),
Address VARCHAR(100),
Phone VARCHAR(15),

```
Gender VARCHAR(10)
);

CREATE TABLE SEMSEC(
SSID VARCHAR(10) PRIMARY KEY,
Sem INT,
Sec VARCHAR(5)
);

CREATE TABLE CLASS(
USN VARCHAR(20),
SSID VARCHAR(10),
PRIMARY KEY (USN, SSID),
FOREIGN KEY (USN) REFERENCES STUDENT(USN),
FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);

CREATE TABLE SUBJECT(
Subcode VARCHAR(10) PRIMARY KEY,
Title VARCHAR(50),
Sem INT,
Credits INT
);

CREATE TABLE IAMARKS(
USN VARCHAR(20),
Subcode VARCHAR(10),
SSID VARCHAR(10),
Test1 INT,
Test2 INT,
Test3 INT,
FinalIA INT,
PRIMARY KEY (USN, Subcode, SSID),
FOREIGN KEY (USN) REFERENCES STUDENT(USN),
FOREIGN KEY (Subcode) REFERENCES SUBJECT(Subcode),
FOREIGN KEY (SSID) REFERENCES SEMSEC(SSID)
);
```

3	21:18:49	CREATE TABLE STUDENT(USN VARCHAR(20) PRIMARY KEY, SName VARCHAR(50), Address VARCHAR(100), Phone VARCHAR(15),...	0 row(s) affected	0.026 sec
4	21:18:49	CREATE TABLE SEMSEC(SSID VARCHAR(10) PRIMARY KEY, Sem INT, Sec VARCHAR(5))	0 row(s) affected	0.0075 sec
5	21:18:49	CREATE TABLE CLASS(USN VARCHAR(20), SSID VARCHAR(10), PRIMARY KEY (USN, SSID), FOREIGN KEY (USN) REFERENCES STU...	0 row(s) affected	0.016 sec
6	21:18:49	CREATE TABLE SUBJECT(Subcode VARCHAR(10) PRIMARY KEY, Title VARCHAR(50), Sem INT, Credits INT)	0 row(s) affected	0.0055 sec
7	21:18:49	CREATE TABLE IAMARKS(USN VARCHAR(20), Subcode VARCHAR(10), SSID VARCHAR(10), Test1 INT, Test2 INT, Test3 INT, FinalIA I...	0 row(s) affected	0.016 sec

PART B – COMPANY DATABASE

STEP 5: Draw ER Diagram



STEP 6: Create Tables

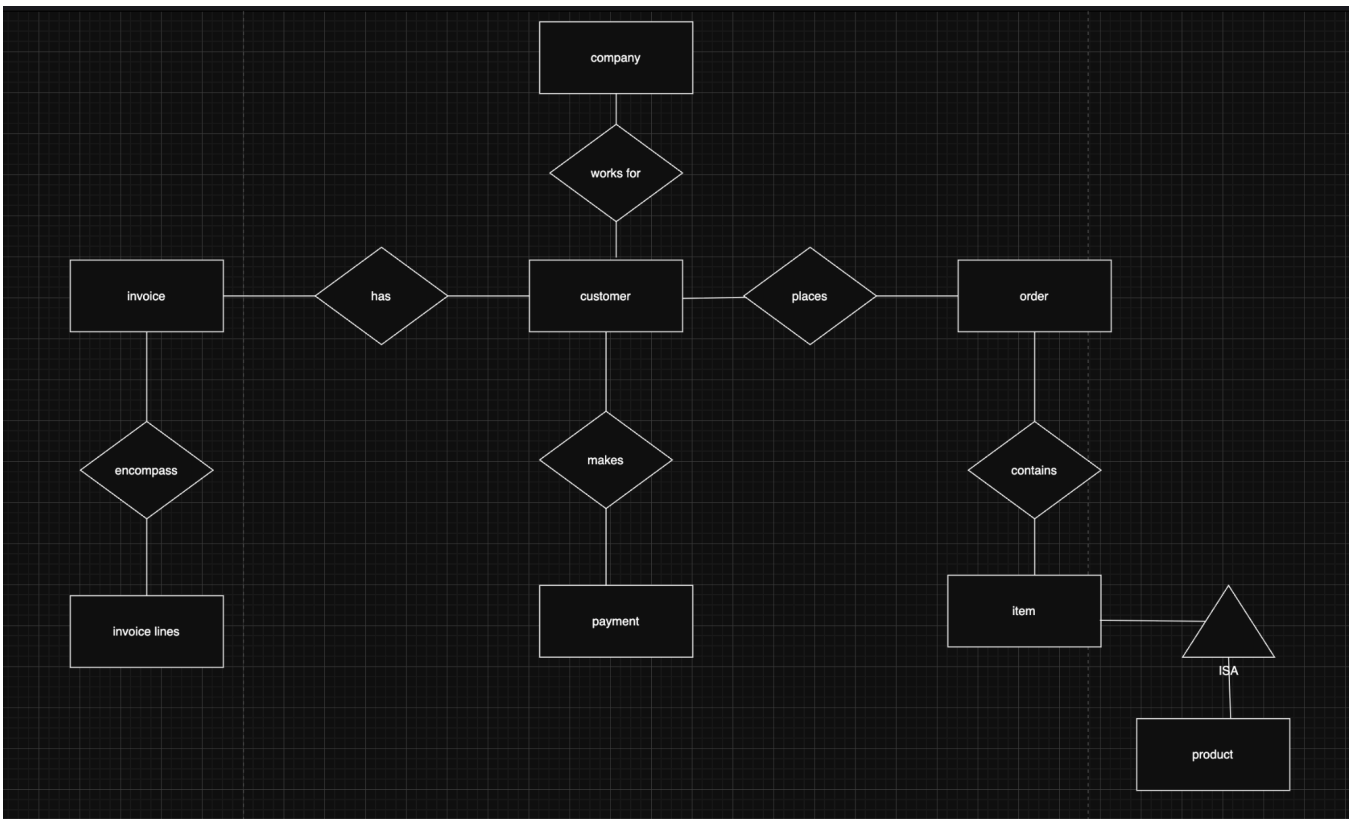
```
CREATE TABLE DEPARTMENT(
Dno INT PRIMARY KEY,
Dname VARCHAR(50),
Mgrssn VARCHAR(20),
Mgrstartdate DATE
);

CREATE TABLE EMPLOYEE(
SSN VARCHAR(20) PRIMARY KEY,
Name VARCHAR(50),
Address VARCHAR(100),
Sex VARCHAR(10),
Salary INT,
Superssn VARCHAR(20),
Dno INT,
FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
```

```
);

CREATE TABLE PROJECT(
PNo INT PRIMARY KEY,
PName VARCHAR(50),
PLocation VARCHAR(50),
DNo INT,
FOREIGN KEY (DNo) REFERENCES DEPARTMENT(Dno)
);

CREATE TABLE WORKS_ON(
SSN VARCHAR(20),
PNo INT,
Hours INT,
PRIMARY KEY (SSN, PNo),
FOREIGN KEY (SSN) REFERENCES EMPLOYEE(SSN),
FOREIGN KEY (PNo) REFERENCES PROJECT(PNo)
);

CREATE TABLE DLOCATION(
Dno INT,
Dloc VARCHAR(50),
FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dno)
);
```

12	00:38:51	CREATE TABLE DLOCATION(Dno INT, Dloc VAR...	0 row(s) affected	0.0060 sec

PART C – CUSTOMER SALE DATABASE

STEP 7: Draw ER Diagram



STEP 8: Create Tables

```
CREATE TABLE CUSTOMER(
Cust_id INT PRIMARY KEY,
cust_name VARCHAR(50)
);

CREATE TABLE ITEM(
item_id INT PRIMARY KEY,
item_name VARCHAR(50),
price INT
);
```

```
CREATE TABLE SALE(
bill_no INT PRIMARY KEY,
bill_date DATE,
cust_id INT,
item_id INT,
qty_sold INT,
FOREIGN KEY (cust_id) REFERENCES CUSTOMER(Cust_id),
FOREIGN KEY (item_id) REFERENCES ITEM(item_id)
);
```

15	01:04:14	CREATE TABLE CUSTOMER(Cust_id INT PRIM...	0 row(s) affected	0.014 sec
16	01:04:14	CREATE TABLE ITEM(item_id INT PRIMARY KE...	0 row(s) affected	0.0047 sec
17	01:04:14	CREATE TABLE SALE(bill_no INT PRIMARY KEY...	0 row(s) affected	0.012 sec

DDL OPERATIONS

STEP 9: Add Email Column

```
ALTER TABLE STUDENT
ADD Email VARCHAR(50);
```

18	01:05:53	ALTER TABLE STUDENT ADD Email VARCHAR(...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.032 sec

STEP 10: Apply NOT NULL

```
ALTER TABLE EMPLOYEE
MODIFY Salary INT NOT NULL;

ALTER TABLE ITEM
MODIFY price INT NOT NULL;
```

20	01:08:19	ALTER TABLE ITEM MODIFY price INT NOT NULL	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.013 sec

STEP 11: Rename Column

```
ALTER TABLE STUDENT
RENAME COLUMN SName TO StudentName;
```

21	01:08:48	ALTER TABLE STUDENT RENAME COLUMN SN...	0 row(s) affected Records: 0 Duplicates: 0 Warnings...	0.013 sec

STEP 12: Drop Table

DROP TABLE DLOCATION;

22	01:09:12	DROP TABLE DLOCATION	0 row(s) affected	0.0085 sec

DML OPERATIONS

STEP 13: Insert Data (Example)

INSERT INTO CUSTOMER VALUES (1,'Rahul');
INSERT INTO CUSTOMER VALUES (5,'Amit');

INSERT INTO ITEM VALUES (101,'Pen',50);
INSERT INTO ITEM VALUES (102,'Bag',800);

INSERT INTO SALE VALUES (1,CURDATE(),5,102,2);

23	01:09:38	INSERT INTO CUSTOMER VALUES (1,'Rahul')	1 row(s) affected	0.0087 sec
24	01:09:38	INSERT INTO CUSTOMER VALUES (5,'Amit')	1 row(s) affected	0.00074 sec
25	01:09:38	INSERT INTO ITEM VALUES (101,'Pen',50)	1 row(s) affected	0.00080 sec
26	01:09:38	INSERT INTO ITEM VALUES (102,'Bag',800)	1 row(s) affected	0.00065 sec
27	01:09:38	INSERT INTO SALE VALUES (1,CURDATE(),5,10...	1 row(s) affected	0.0028 sec

STEP 14: Run Queries

SELECT * FROM CUSTOMER;
SELECT S.bill_no, C.cust_name, I.item_name
FROM SALE S
JOIN CUSTOMER C ON S.cust_id=C.Cust_id
JOIN ITEM I ON S.item_id=I.item_id;

100% 1:7

Result Grid Filter Rows: Search Export:

bill_no	cust_name	item_name
1	Amit	Bag

CUSTOMER 1 Result 2 Read Only

Action Output

	Time	Action	Response	Duration / Fetch Time
25	01:09:38	INSERT INTO ITEM VALUES (101,'Pen',50)	1 row(s) affected	0.00080 sec
26	01:09:38	INSERT INTO ITEM VALUES (102,'Bag',800)	1 row(s) affected	0.00065 sec
27	01:09:38	INSERT INTO SALE VALUES (1,CURDATE(),5,102,2)	1 row(s) affected	0.0028 sec
28	01:10:07	SELECT * FROM CUSTOMER LIMIT 0, 1000	2 row(s) returned	0.0074 sec / 0.00034...
29	01:10:07	SELECT S.bill_no, C.cust_name, I.item_name FROM SALE S JOIN CUSTOMER C ON S.cust_id=C.Cust_id JOIN ITEM I ON S.item_id=I...	1 row(s) returned	0.00060 sec / 0.000...

CONCLUSION:

In this lab experiment, we learned how to design ER diagrams and convert them into relational tables.We understood how to apply primary keys and foreign keys.We also practiced DDL and DML commands in SQL.
This experiment improved our understanding of database design and implementation.